SOFTWARE ENGINEERING LABORATORY

LAB–1

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

LIBRARY MANAGEMENT SYSTEM

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the Library Management System (LMS). The system aims to automate the daily activities of a library, including book management, student registration, book issuing, returning, searching, and fine calculation.

This document provides a clear understanding of the system's requirements for developers, testers, administrators, librarians, and other stakeholders.

1.2 Scope

The Library Management System is a software application designed to simplify and automate library operations. The system will allow:

- Students to search for available books.
- Librarians to issue and return books.
- Administrators to manage books and user accounts.
- The system to calculate overdue fines automatically.
- Users to view their borrowing history.
- Librarians to generate reports related to books and transactions.

The system will reduce manual paperwork, improve accuracy, save time, and make library operations more efficient.

1.3 Definitions, Acronyms and Abbreviations

Term	Description
LMS	Library Management System
SRS	Software Requirements Specification
Admin	System Administrator
User	Student or library member
ISBN	International Standard Book Number
DBMS	Database Management System
UI	User Interface
Fine	Amount charged for late book returns

1.4 References

- IEEE Std 830-1998 – IEEE Recommended Practice for Software Requirements Specifications.
- Software Engineering – Ian Sommerville.
- Software Engineering – Roger S. Pressman.

1.5 Overview

This document describes the overall structure and requirements of the Library Management System. Section 1 introduces the project, its purpose, scope, and terminology. Section 2 describes the overall system, its users, operating environment, and constraints. Section 3 specifies the functional and non-functional requirements of the system.

2. Overall Description

2.1 Product Perspective

The Library Management System is a standalone software application that maintains library-related information in a centralized database. It replaces the traditional manual library management process with a computerized system.

The system consists of three major components:

- **User Interface:** Allows students, librarians, and administrators to interact with the application.
- **Application Logic:** Handles book issuing, returning, searching, and fine calculations.
- **Database:** Stores information about books, students, librarians, and transactions.

2.2 Product Functions

- User registration and login.
- Book addition, deletion, and modification.
- Searching books by title, author, category, or ISBN.
- Issuing books to registered students.
- Returning issued books.
- Automatic fine calculation for overdue books.
- Maintaining student borrowing records.
- Generating library reports.
- Managing book availability.

2.3 User Classes and Characteristics

1. Administrator

- Manages librarian accounts.
- Manages system settings.
- Monitors overall library activities.
- Has access to all system functionalities.

2. Librarian

- Adds, updates, and removes book records.
- Issues books to students.
- Accepts returned books.
- Calculates and manages fines.
- Generates reports.

3. Student

- Logs into the system.
- Searches for books.
- Checks book availability.
- Views currently issued books.
- Checks borrowing history and pending fines.

2.4 Operating Environment

- Operating System: Windows, Linux, or macOS.
- Frontend: HTML, CSS, JavaScript.
- Backend: Node.js and Express.js.
- Database: MongoDB.
- Browser: Google Chrome, Mozilla Firefox, or Microsoft Edge.
- Internet Connection: Required for accessing the web-based application.

2.5 Design and Implementation Constraints

- Only registered users can access protected functionalities.
- Students cannot directly modify book records.
- Only librarians and administrators can manage books.
- A book cannot be issued if no copy is available.
- The system must maintain accurate transaction records.
- The application must protect user credentials and personal information.

2.6 Assumptions and Dependencies

- Every student has a unique library membership ID.
- Each book has a unique ISBN or book identification number.
- Librarians are responsible for maintaining accurate book records.
- Users have access to a computer or smartphone with an internet connection.
- The database server must be available for normal system operation.

3. Specific Requirements

3.1 Functional Requirements

Functional requirements describe the operations that the Library Management System must perform.

FR-1: User Registration

- The system shall allow new students to register.
- The system shall collect name, email, student ID, and password.
- The system shall prevent duplicate registrations using the same student ID or email.

FR-2: User Authentication

- The system shall allow registered users to log in using their credentials.
- The system shall validate the entered username and password.
- The system shall provide access according to the user's role.
- The system shall allow users to log out securely.

FR-3: Book Management

- The system shall allow librarians to add new books.
- The system shall allow librarians to update book details.
- The system shall allow librarians to delete book records when appropriate.

- The system shall maintain book title, author, ISBN, category, and quantity.

FR-4: Book Searching

- The system shall allow students and librarians to search for books.
- Users shall be able to search using book title, author, ISBN, or category.
- The system shall display book availability status.

FR-5: Book Issuing

- The system shall allow librarians to issue books to registered students.
- The system shall verify book availability before issuing.
- The system shall record the issue date and due date.
- The system shall update the available book quantity after issuing.

FR-6: Book Returning

- The system shall allow librarians to record returned books.
- The system shall update book availability after a successful return.
- The system shall calculate overdue fines if the book is returned after the due date.

FR-7: Fine Management

- The system shall calculate fines based on the number of overdue days.
- The system shall display pending fines to students.
- The system shall allow librarians to record fine payments.

FR-8: Borrowing History

- The system shall maintain the borrowing history of each student.
- Students shall be able to view their currently issued books.
- Students shall be able to view previously returned books.

FR-9: Report Generation

- The system shall generate reports of issued books.
- The system shall display available and unavailable books.
- The system shall generate overdue book reports.
- The system shall provide information about total books and registered users.

3.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance expectations of the system.

1. Performance

- The system should display search results within 3 seconds under normal operating conditions.
- The system should support multiple users simultaneously.
- Book issuing and returning operations should be processed efficiently.

2. Security

- User passwords must be stored securely using password hashing.
- The system shall provide role-based access control.

- Unauthorized users must not access administrative functions.
- User sessions must be handled securely.

3. Reliability

- The system should maintain accurate book and transaction records.
- The system should prevent duplicate or inconsistent transactions.
- Regular database backups should be maintained.

4. Usability

- The user interface should be simple and user-friendly.
- Navigation should be easy for students and librarians.
- Error messages should be clear and understandable.

5. Availability

- The system should be accessible whenever the library is operational.
- Database failures should be handled without corrupting existing records.

6. Maintainability

- The application should use modular programming.
- The system should be easy to update and modify.
- Errors should be logged for debugging and maintenance.

7. Scalability

- The system should support an increasing number of students and books.
- The database should be capable of handling growing library records.

3.3 External Interface Requirements

3.3.1 User Interface

- Login and registration page.
- Student dashboard.
- Librarian dashboard.
- Administrator dashboard.
- Book search page.
- Book management page.
- Book issue and return page.
- Fine and report management page.

3.3.2 Hardware Interface

- Desktop computers or laptops.
- Smartphones and tablets.
- Server hardware for hosting the application and database.

3.3.3 Software Interface

- Frontend browser interface.

- Node.js and Express.js backend.
- MongoDB database.
- Operating system and web browser.

3.3.4 Communication Interface

- The application shall communicate using HTTP/HTTPS protocols.
- Data exchange between frontend and backend shall use REST APIs.
- HTTPS shall be used to secure communication.

3.4 Database Requirements

The system shall maintain the following major collections/tables:

Collection / Table	Required Attributes
Student	Student ID, Name, Email, Password, Contact Number
Book	Book ID, ISBN, Title, Author, Category, Total Quantity, Available Quantity
Librarian	Librarian ID, Name, Email, Password
Issue Record	Issue ID, Student ID, Book ID, Issue Date, Due Date, Return Date, Status
Fine	Fine ID, Student ID, Issue ID, Fine Amount, Payment Status

4. System Constraints

- Only authorized users can access the system.
- A student cannot issue more than the permitted number of books.
- A book cannot be issued when its available quantity is zero.
- Each transaction must have a unique identification number.
- The system depends on database availability.
- Fine calculations must follow the library's predefined rules.

5. Acceptance Criteria

The Library Management System will be considered successful when:

- Students can register and log in successfully.
- Librarians can add, update, and delete book records.
- Users can search for books and view their availability.
- Librarians can issue and accept returned books.
- The system calculates overdue fines correctly.
- Students can view their borrowing history.
- Administrators can generate reports.
- Unauthorized users cannot access restricted functionalities.
- All important records are stored and retrieved correctly.

6. Conclusion

The Library Management System provides an efficient and organized solution for managing library operations. It reduces manual work, minimizes errors, saves time, and improves the overall experience of students and librarians.

This SRS document defines the functional requirements, non-functional requirements, system constraints, interfaces, and database requirements of the proposed system. It will serve as a foundation for the design, development, testing, and maintenance of the Library Management System.

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